

1.

```
if (grade >= 90) {  
    System.out.println("Great Job!");  
}  
if (number < 20 || number > 50) {  
    System.out.println("Error");  
}  
if (y < 100) {  
    y += 2; // Equivalent to: y = y + 2;  
}
```

2.

```
if (num1 > num2) {  
    System.out.println("first number is larger");  
} else if (num2 > num1) {  
    System.out.println("second number is larger");  
} else {  
    System.out.println("numbers are equal");  
}
```

3.

```
a) If (num % 2 == 0) {  
    System.out.println("even number");  
} else {  
    System.out.println("odd number");  
}  
b) switch (num % 2) {  
    case 0:  
        System.out.println("even number");  
        break;  
    default:  
        System.out.println("odd number");  
}
```

4.

```
int randomInt1_50 = (int) (Math.random() * 50) + 1;
```

```
int randomInt20_100 = (int) (Math.random() * 81) + 20;
```

```
double randomDouble10_20 = Math.random() * 10.0 + 10.0;
```

5.

If the user enter 18 and 65 the code will break because they are not defined

6.

- a) True
- b) False
- c) False
- d) False
- e) true
- f) True
- g) False

8.

- a) False
- b) False
- c) False
- d) True
- e) False
- f) True
- g) True
- h) False
- i) True
- j) true